VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)

DEPARTMENT OF INFORMATION TECHNOLOGY

Letter of acceptance for UG / PG Mini Projects

Name of the Department: INFORMATION TECHNOLOGY	**ROLL NUMBERS**
Name of the Guide:	1. M. Vaishnavi (23071A1238)
Project Title: Speech Emotion Recognition Using CNN–LSTM for Mental Health Monitoring	2. R. Nihesh (23071A1250)
Section: A	3. R. Laxmi Prasanna (23071A1253)
Submission Date:	4. R. Sri Koushik (23071A1254)

Project Abstract:

This project focuses on automatic recognition of human emotions from speech signals using deep learning techniques. A hybrid **CNN-LSTM model** is used to extract spatial and temporal features from audio signals. Speech features such as **MFCCs** are extracted and processed using CNN for feature learning and LSTM for sequence modeling. The model is trained on the **RAVDESS dataset** to classify emotions like happy, sad, angry, calm, and neutral. The system aims to support **mental health monitoring** by identifying emotional patterns from speech in an automated and non-intrusive manner.

Problem Definition:

Mental health conditions often influence speech and emotional expression. Manual emotion assessment is subjective and not scalable. The problem is to design an automated system that accurately identifies emotions from speech signals using deep learning models for effective mental health monitoring.

Requirements:

Hardware: PC/Laptop with minimum 8GB RAM
Software: Python, TensorFlow/Keras, Librosa, NumPy, Scikit-learn
Dataset: RAVDESS Emotional Speech Dataset

Future Expansion of the Project:

- Real-time emotion detection from live speech
- Multilingual emotion recognition
- Integration with web or mobile mental health applications
- Multimodal emotion analysis using speech and facial expressions

Modifications / Suggestions given (if any):	
Signature of HOD:	Signature of the Project Mentor